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Stability and *Collapse* of Pollination Networks

A Network Science Perspective • Indian Institute of Technology Bhilai

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Networks	**172**
Continents	**6**
GNN AUC	**0.960**
Species	**11.5K**

Abstract: Pollination networks sustain **about 88% of flowering plants** and **75% of global food crops**. We analyse **172 real-world bipartite networks** across 6 continents using cascading co-extinction simulation, climate stress modelling, adaptive rewiring, early warning signal detection, Random Forest extinction risk prediction, **GraphSAGE-based role discovery**, and species transplantation to characterise structural vulnerability and collapse dynamics at continent scale.

1. Dataset & Network Model

Bipartite graph $G = (U \cup V, E)$: pollinators (U), plants (V). **172 city-level datasets** from Web of Life were grouped into 6 continental meta-networks for macro-scale comparison, while all 172 were retained for network-level GNN analysis.

- **Degree centrality**: generalist identification
- **Betweenness centrality**: bridge species
- **Eigenvector centrality**: systemic influence
- **NODF nestedness**: specialist-generalist hierarchy
- **Modularity** Q: community compartmentalisation

2. Continental Network Properties

Continent	Plants	Pollin.	NODF	Q	Conn.
Africa	147	737	18.2	0.453	0.15
Asia	543	2947	4.3	0.602	0.12
Europe	417	1988	5.4	0.627	0.13
N. America	909	1855	3.2	0.270	0.09
Oceania	127	299	9.4	0.488	0.16
S. America	1089	907	3.9	0.773	0.08

South America ($Q = 0.773$) is most modular; **North America** ($Q = 0.270$) is most integrated and most vulnerable to targeted attack.

3. GNN Functional Role Discovery

Two-layer **GraphSAGE** on all 172 networks. Link-prediction pre-training → k-means ($k = 4$) → Generalist / Specialist / Connector / Keystone.

- Connector: **38%**
- Specialist: **37%**
- Keystone: **15%**
- Generalist: **10%**

Mean GNN AUC: 0.960 (171/172 networks ≥ 0.85). **Mean silhouette: 0.384** ($k = 4$). Connectance-generalism correlation: $r = 0.48$, $p < 0.001$.

4. Cascading Co-extinction

Pollinators removed sequentially; plants co-extinct when all links are lost. 50 repeats for random-removal average.

- Targeted-removal half-life near **20%** pollinator loss
- Random-removal half-life near **65%** pollinator loss
- **2.5x to 4x** difference in network half-life

North America ($Q = 0.270$) collapses earliest; targeted removal generates 80–300 isolated components.

5. Climate Stress Simulation

Climate stress is modelled as gradual reduction in interaction strength ($s : 0 \to 1$). All networks collapse rapidly beyond $s \approx 0.75$–0.85.

- S. America threshold: about 0.85
- Asia threshold: about 0.85
- Europe threshold: about 0.80
- N. America threshold: about 0.75

Pollinators decline before plants across all networks; pollinator monitoring provides earlier stress warning.

6. Early Warning Signal Detection

Critical Slowing Down (CSD): rising variance and lag-1 autocorrelation precede collapse bifurcation.

- CSD autocorrelation threshold: 0.9
- Variance rises near stress ≈ 0.70–0.75
- Warning window: **10 to 30 years** before transition
- Oceania shows earliest warning (limited island redundancy)

7. Adaptive Rewiring ($p = 0.6$)

Surviving plants seek alternative pollinators proportional to degree when a partner is lost. Resilience gain is the area between rewiring and no-rewiring trajectories.

- Largest gain: Europe and Oceania (high nestedness)
- Smallest gain: Asia and N. America (modular, locally saturated)
- Rewiring benefit is positively correlated with nestedness

Conservation implication: habitat corridors enabling novel encounters can improve resilience beyond species-level protection.

8. ML Extinction Risk Prediction

Random Forest classifier (100 trees, 5-fold CV) on node-structural features.

- Eigenvector centrality: about 40% importance
- Degree centrality: about 20%
- Betweenness centrality: about 15%
- Module connectivity: about 10%

Eigenvector centrality consistently outperforms raw degree for extinction risk prediction.

9. Species Transplantation

Introduced `Unidentified sp16 M_PL_050` into target community M_PL_050 (17 plants × 51 pollinators).

Metric	Before	After	Δ
Interactions	129	132	+3
NODF	38.94	38.06	-0.88
Modularity Q	0.4627	0.4502	-0.013
LCC size	68	69	+1

Key constraint: transplanted species should target bridge plants (high betweenness), not already over-connected hubs.

Key Findings & Conclusions

01. Continental Architecture Diversity

South America ($Q = 0.773$) is most stress-resistant. **North America** ($Q = 0.270$) is most vulnerable, with 2.5x–4x faster collapse under targeted versus random removal. Africa's high nestedness (NODF=18.2) drives synchronous tipping-point collapse.

02. GNN Corpus & Eigenvector Dominance

Across 172 networks, **Connector (38%)** and **Specialist (37%)** are co-dominant regimes. **Eigenvector centrality** outperforms raw degree as extinction-risk predictor.

03. Early Warning & Rewiring Buffer

CSD signals are detectable **10–30 years** before collapse. **Adaptive rewiring** improves resilience across all networks; managed transplantation helps when species bridge structural gaps.

Reference

J. Bascompte, P. Jordano, C. J. Melian, and J. M. Olesen, "The nested assembly of plant-animal mutualistic networks," *Proc. Natl. Acad. Sci. U.S.A.*, vol. 100, no. 16, pp. 9383–9387, 2003.

Data source: Web of Life (`https://www.web-of-life.es`)

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